DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2016
YEAR SIX
H2 BIOLOGY (9648)
PAPER 2

Structured Questions Answers

Question 1

(a)
Phospholipids arranged in a **monolayer** with **hydrophobic hydrocarbon tails face inward** to interact with triglycerides in interior and **hydrophilic phosphate heads face outwards** to interact with the aqueous medium;

Makes hydrophobic lipids soluble for transport in the blood;

(b)(i)
A: Golgi apparatus;
B: Rough endoplasmic reticulum; (A: ribosome, R: ribosomes)

(b)(ii)
A:
chemically modifies the LDL receptors (e.g. glycosylates);
packages LDL receptors the into transport vesicles to be **targeted** to the plasma membrane for insertion into membrane;

B: RER
Ref to **translation** of LDL receptor mRNA at bound **ribosome**s to from LDL receptor polypeptide;
folding of polypeptide into native configuration inside cisternal space;
transport of newly synthesized LDL receptor to the cis face of the golgi apparatus within transport vesicles;
2 max

OR
B: ribosome
Ref to **translation** of LDL receptor **mRNA** to from LDL receptor **polypeptide**;
Ref to **peptidyl transferase** in large ribosomal subunit catalyzing the formation of **peptide bonds** between amino acids;

(c)
Fluid nature of phospholipids moving in the membrane allows invagination of plasma membrane/ fusion of two ends of plasma membrane to form endocytic vesicle;
Mosiac – membrane has proteins embedded such as the LDL receptor. Binding of LDL to receptor triggers invagination of the membrane;

Question 2

(a)(i) **Tyrosine (tyr);**

(a)(ii) **Glycine (gly);**

(b)(i) Increase coding capacity; AVP

(b)(ii)

Ribosomes will continue translation as **STOP codon now codes for an amino acid**;

Translation of the mRNA continues into the **3'-untranslated region**;

Translation continues until the **next in-frame stop codon downstream**;

2 max

(b)(iii)

Nonstop mutations differ from nonsense mutations in that they **do not create a stop codon but, instead, delete one**;

(c)(i)

Deletion of CTT (or TCT), at corresponding amino acid positions 507 and 508;

Change of codon from ATC to ATT results in **same amino acid encoded at position 507, isoleucine**;

Change of codon from TTT to GGT results in a **different amino acid at position 508 - glycine instead of phenylalanine** (OR **deletion of phenylalanine at position 508**);

Overall **primary structure of the polypeptide / amino acid sequence is changed;**

Ref. to change **affecting folding of the polypeptide / incorrect folding** of the polypeptide;

4 max

(c)(ii)

Single base substitution, at corresponding **amino acid position 523**, where the third base, **C is replaced by A**, changing the codon UGC to a **stop codon** UGA;

Results in **premature termination of translation**, producing a **truncated / shortened polypeptide chain**;

Question 3

(a)
Icosahedral capsid head;
protects the viral ds DNA when virus is outside its host;

(b)
Contractile sheath punctures a hole through bacterial host cell wall to allow entry of viral DNA into the host cell;
Ref to gp19 tube / gp27 – gp5 spike;
As viral DNA too large and charged to cross bacterial cell wall and cell membrane on its own;
2 max

(c)
Phage uses host cell DNA polymerase to synthesize new copies of phage DNA;
And host RNA polymerase and ribosomes to synthesize new phage proteins;
new phage particles then assemble from newly synthesized components;
Phage coded lysozyme breaks down peptidoglycan cell wall resulting in osmotic lysis of host and release of phage particles;
4 max

(d)
During T4 phage replication, a phage enzyme degrades bacterium host cell's DNA;
During assembly of the virus, a fragment of the host's genome may be packaged inside the virus capsid by mistake;
The resultant defective phage then injects this bacterial gene into another bacterium, thus resulting in horizontal gene transfer between bacteria;
2 max

Question 4

(a)(i)

G_1 checkpoint (**2 max**): checks that

- sufficient nutrients present;
- environment is favourable / need for new cells for replacement;
- sufficient growth of the cell / cell reach a minimum size;
- sufficient organelles;
- DNA not damaged and can be replicated;
- growth factors are present;

M checkpoint (**2 max**):

- Checks for attachment of spindle fibers to the kinetochores (centromeres) of the chromosomes;
- Ensures correct alignment of chromosomes at metaphase plate;
- Allows separation of sister chromatids equally at anaphase;

(a)(ii)

Synthesis of proteins/RNA/enzymes;

Formation of new organelles;

ATP production;

(a)(iii)

Quiescence is **reversible** while senescence is **irreversible**;

Quiescence occurs when cells are **neither dividing nor preparing to divide** (e.g. when they ‘exit’ from the cell cycle) while senescence occurs **in response to DNA damage or degradation** that would make a cell's progeny nonviable / when cells reach the **Hayflick limit** (i.e. reproductive limit);

(b)(i)

<u>96 telomeres</u>;

There are 11 tetrads/ bivalents and 1 pair of unpaired chromosomes (**A!** any suitable number based on clarity of diagram). Each chromosome consists of two sister chromatids, hence total number of sister chromatids is 48. Both ends of each sister chromatid is flanked by telomeres, hence the total number of telomeres is 96;

(b)(ii)

- Centromeres **hold genetically identical sister chromatids together** as the chromosomes align themselves at the metaphase plate;
- **Kinetochore proteins** bind to the centromeres;
- During metaphase spindle fibres from both poles attach to the kinetochore proteins;
- During anaphase, shortening of the spindle fibres and the duplication of the centromeres results in sister chromatids being separated to opposite poles;
- Such that at the end of mitosis, each daughter nuclei contains the identical genetic material.

2 max

Question 5

(a)(i)

Parental phenotypes **malvidin producing plant X non-producing**
Parental genotypes **KKdd X k'k'DD**

Gametes **Kd X k'D** [1]

F1 phenotype **Non-producing plant X Non-producing plant**
F1 genotype **Kk'Dd X Kk'Dd**

F1 gametes **KD k'D Kd k'd X KD k'D Kd k'd**
[1]

F2 genotype

	KD	k'D	Kd	k'd
KD	KKDD Non-producing	Kk'DD Non-producing	KKDd Non-producing	Kk'Dd Non-producing
k'D	Kk'DD Non-producing	k'k'DD Non-producing	Kk'Dd Non-producing	k'k'Dd Non-producing
Kd	KKDd Non-producing	Kk'Dd Non-producing	KKdd Malvidin producing	Kk'dd Malvidin producing
k'd	Kk'Dd Non-producing	k'k'Dd Non-producing	Kk'dd Malvidin producing	k'k'dd Non-producing

[1]

F2 genotype:	**KKDD, Kk'DD, KKDd, Kk'Dd, k'k'Dd, k'k'DD, k'k'dd**	**KKdd, Kk'dd**
F2 phenotype:	**non-producing**	**malvidin producing plant**
F2 phenotypic ratio:	**13**	**: 3**

[1]

(a)(ii)
<u>Allele D is epistatic</u> over the <u>K and k' locus</u> where it <u>prevents the formation of malvidin</u>;
By synthesizing a gene product which suppresses the action of the enzyme encoded by K /
D codes for an inhibitor which binds to and prevents the action of the enzyme encoded by K;

(b)(i)

$$\chi^2 = \frac{(40-30)^2}{30} + \frac{(42-30)^2}{30} + \frac{(20-30)^2}{30} + \frac{(18-30)^2}{30}$$

= 16.27; (2dp);

(b)(ii)
At n=3 where the calculated value of χ^2 (16.27), the corresponding probability is 0.001;

Critical χ^2 value (7.82) < calculated χ^2 value (16.27), therefore there is a significant difference between observed and expected values;

The cross does not follow ratio of 1:1:1:1. Any difference is not due to chance alone but other factors must be at operation (eg. linked genes);

Question 6

(a)(i)

Yeast are still respiring aerobically, using up oxygen in jar after it was sealed;

(a)(ii)

Anaerobic respiration - Ref. to incomplete/partial oxidation of glucose during glycolysis in the absence of oxygen;

1 molecule of glucose is broken down to 2 molecules of pyruvate with the net yield of 2 ATP by substrate level phosphorylation and 2 reduced NAD/ 2NADH;

Alcoholic fermentation occurs to regenerate NAD^+ for glycolysis to continue;

Pyruvate is decarboxylated to form acetaldehyde with the removal of carbon dioxide;

Acetaldehyde is then reduced to ethanol by accepting H atoms from NADH to regenerate NAD^+;

Resulting in an increase in concentration of alcohol from 0 to 16% from 24 to 96 hr;

(a)(iii)

Yeast is killed at 16% ethanol;

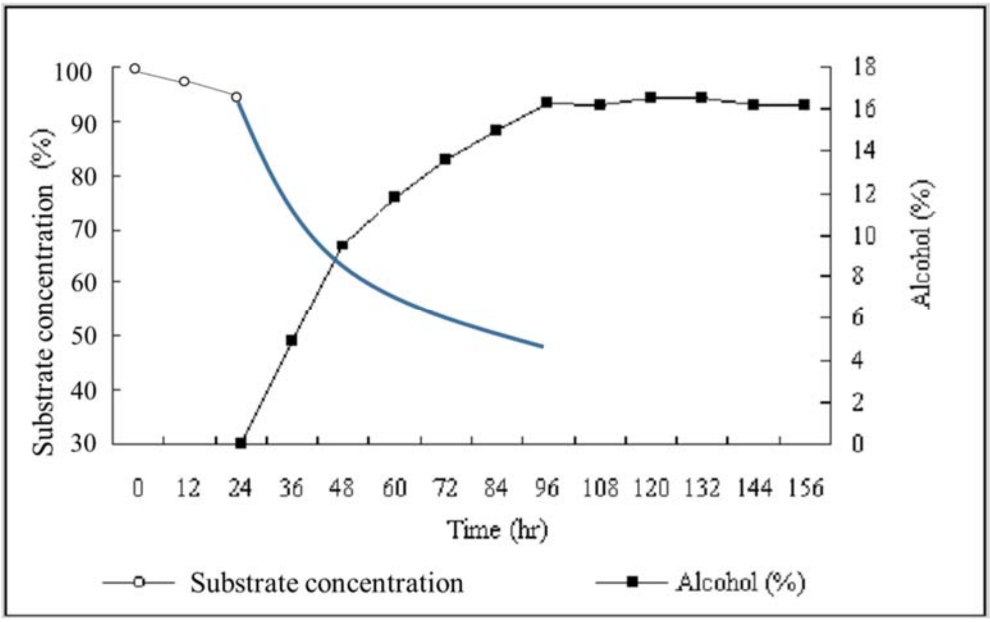
ethanol is organic and thus dissolves and disrupts phospholipid bilayer of the cell membranes in yeast;

enzymes in yeast are also denatured by the high ethanol concentration;

2 max

(b)

Curve with sharper initial decrease in substrate concentration than original gradient from 0-24h;



Question 7

(a)(i)
GABA binds to receptors and thus leads to the opening of K+ / Cl- gated channels;
This results in more Cl- diffusing into / more K+ diffusing out of postsynaptic membrane, thus membrane potential becomes more negative / hyperpolarized;

(ii)
Acetylchloline from P will cause depolarisation while GABA from Q will cause hyperpolarisation / ref. to acetylcholine causing influx of Na+ & GABA causing influx of Cl- and efflux of K+;
Ref. to spatial summation of these potentials counteracts each other resulting in a weak stimulus in neurone B;

(iii)
Barbiturates bind to GABA receptor and cause more influx of Cl- ions and efflux of K+ ions;
Prevents enzyme from degrading GABA;
Prevents GABA from unbinding from GABA receptor;
Ref. to binds to GABA and helps GABA bind to the GABA receptor;
Ref. to barbiturates causes increased secretion of GABA;

1 Max.

(b)
Ref. to the summation of EPSPs produced by repeated stimulation of only ONE presynaptic neurone (at high frequency);
If more neurotransmitters are released into the cleft before the first EPSP is destroyed;
Additive effects of several EPSPs may exceed threshold potential to result in an action potential in the post synaptic neurone;

2 Max.

Question 8

(a)(i)
Compare DNA sequence of a particular common gene between different species of fish / comparison/alignment of homologous genetic sequences;
% sequence homology indicates degree of evolutionary closeness / number of mutation in genetic sequence is used to calculate the length of time since divergence;
Neutral mutations are accumulated at a relatively constant rate and use as a molecular clock;

2 Max.

(a)(ii)
Quantifiable where protein, nucleic acid sequence data are precise and accurate and easy to quantify / convertible to numerical form for mathematical and statistical analysis**;**
Based on the idea of molecular clock, that the rate of mutation is constant, species can be arranged in order of time of evolution;
Objective where data is based strictly on heritable material / can be easily described in an unambiguous manner / some morphological similiarities may be analogous / ref. convergent evolution**;**
Use of phenotypically non-visible characteristics / considers changes caused by silent mutation which is not shown on the phenotype;

2 Max.

(a)(iii)
A. taeniatus and *A. virginicus* are closely related where they share a (recent) common ancestor. They have a high percentage homology in DNA sequence alignment;

(b)
Geographical isolation /Isthmus of Panama is a physical barrier / ref. allopatric speciation;
Disruption to gene flow in the ancestral population where there is no interbreeding between the organisms in the Pacific Ocean and Caribbean Sea;
Genetic variations exist within each sub-population due to mutation or genetic recombination;
Different selection pressures in Pacific Ocean and Caribbean Sea. List 1 eg. food availability/ salinity /temperature/ different predators;
Individuals with traits that are selectively advantageous in the particular environment survive, reproduce and pass on their alleles to offspring;
There will be changes in allele frequency of gene pool and accumulation of genetic changes takes place over many generations;
Speciation into *A. taeniatus* and *A. virginicus* takes place when the two populations ultimately cannot interbreed to produce viable, fertile offspring;

4 Max.

Essay Answers

Essay Question 1

(a) Describe the protein folding of an enzyme and relate its structure to its function. [10]

1. Enzymes are globular proteins with unique three-dimensional conformation / tertiary / quaternary structure;
2. Ref to primary structure being the unique sequence and number of amino acids in a polypeptide linked by peptide bonds;
3. Ref to secondary structure being the regular coiling and folding/pleating of the polypeptide held by *hydrogen bonds** between CO and NH groups of the peptide bonds / polypeptide backbone;
4. In *alpha helix**, *hydrogen bonds** form between CO and NH groups 4 a.a. apart, forming a 3D helical structure
5. In *beta pleated sheet**, *hydrogen bonds** form between CO (or NH) group of one region/segment and NH (or CO) group of an adjacent region/segment of a single polypeptide chain, forming a flat/pleated sheet;
6. Tertiary structure refers to the folding of polypeptide into a specific conformation, held by bonds between *R-groups** of structural amino acids within same polypeptide, maintained by *hydrophobic interaction, hydrogen bonds, ionic bonds, disulfide bridges;*
7. Ref to quaternary structure: more than 1 polypeptide chain to form functional protein held by *hydrophobic interaction, hydrogen bonds, ionic bonds, disulfide bridges* between R groups between polypeptide chains;
8. Folding gives rise to a specific cleft / groove - *active site* that is *complementary in shape and charge* to its *substrate*.
9. Folding brings catalytic amino acids and binding amino acids far apart in the primary structure / polypeptide close together in the active site
10. R groups of binding residues bind reversibly with substrate to position it in the correct orientation for catalysis to occur.
11. R groups of catalytic residues present within active site catalyze conversion of substrate to product.
12. The rest of the amino acids in the protein molecule are structural residues - provides a framework to maintain active site configuration
13. Active site may not be a rigid receptacle → ref to induced fit model – entrance of substrate induces enzyme to change its shape slightly to 'wrap around' substrate, bringing R group of active site into positions that enhance their ability to catalyze the chemical reaction;
14. Some enzymes contain another site (apart from active site) for another molecule to bind to (cofactors/allosteric molecules) allowing for regulation of enzyme activity;
15. Enzymes are soluble due to arrangement of hydrophilic residues on the surface and hydrophobic residues in the interior, allowing them to catalyse reactions in the aqueous environment of the cell;

(b) Describe the effect of pH on enzymes and their activity. [4]

1. Reference of influence of pH to the effect on ionic bonds, hydrogen bonds;
2. Change in charges of catalytic residues in active site affects catalytic function of enzyme;
3. Changes in shape / conformation and configuration of the tertiary structure affecting the fit and binding of the substrate to the active site;
4. Optimum pH / show graph of narrow pH range / named example;
5. Denaturation results in loss of structure and activity;
6. Denaturation is often reversible, restoring pH to optimum restores enzyme activity;

(c) Compare and contrast competitive and non-competitive inhibitors and their effects on the rate of enzyme activity. [6]

1 Both serve to lower rate of enzyme activity by preventing substrate from binding to active site
2 Both involve reversible binding of inhibitor to enzyme

	Competitive	**Non-competitive**
3 Structure	Structural similarity to substrate	No structural similarity to substrate
4 Binding site	Binds to active site	Binds to site other than active site – allosteric site
5 Competing for active site	Competes with substrate for active site	Does not compete with substrate for active site
6 Conformation of Enzyme	Does not change conformation of enzyme upon binding	Changes conformation of enzyme upon binding such that substrate can no longer bind to active site
7 Effect of increasing [S]	Increasing [S] concentration alleviates effect of inhibitor	Increasing [S] concentration does not alleviate effect of inhibitor
8 V_{max}	Max rate of reaction can be reached (with increased S concentration) / V_{max} unchanged	Max rate of reaction cannot be reached (with increased S concentration) / V_{max} reduced
If points 5 and 6 not awarded, can award point 9		
9 Mode of action	Competes with substrate for active site	Changes conformation of enzyme upon binding such that substrate can no longer bind to active site

Essay Question 2

(a)

Marking Point		Krebs cycle	Calvin cycle
1	**Location**	Mitochondrial matrix	Chloroplast stroma
2	**Substrate**	Acetyl-CoA and oxaloacetate combines to form citrate	CO_2 and Ribulose bisphosphate (RuBP)
3	**Products**	Each glucose molecule gives rise to: 6 NADH 2 $FADH_2$ 2 ATP 4 CO_2	For every 3 molecules of CO_2 that enter the cycle, one triose phosphate / Glyceraldehyde 3 phosphate is made
4	**Regenerated / Starting material**	Oxaloacetate is the starting material that is eventually regenerated	Ribulose bisphosphate (RuBP) is the starting material that is eventually regenerated
5	**ATP**	Produced via substrate level phosphorylation	Used in reduction of glycerate-3-phosphate where energy Is required through hydrolysis of ATP
6	**Electron carriers / donors**	Use NAD^+ and FAD for the oxidation of the intermediates of the cycle by serving as electron acceptors	Uses NADPH / reduced $NADP^+$ to reduce glycerate-3-phosphate to triose phosphate by serving as electron donors
7	**Overall**	Catabolic	Anabolic
8	**Role of CO_2**	CO_2 is released as a result of decarboxylation reactions	Required for carbon fixation. CO_2 is used to convert Ribulose bisphosphate (RuBP) to form an unstable 6C compound that breaks down to form glycerate-3-phosphate
9	**Role of O_2**	Occurs only when O_2 is present	Does not require O_2

8 Max.

(b)

Similarities

1. Both have electron carriers embedded in membranes - inner membrane of mitochondrion and thylakoid membrane of chloroplast;
2. Both involve electrons being passed down a series of electron carriers with increasing electronegativity and in order of decreasing energy levels;
3. Energy released from electron transport chain is used to generate a proton gradient / proton motive force;
4. Both involves diffusion of protons down a concentration gradient through ATP synthase / ref. chemiosmosis;
5. Potential energy of the proton gradient is used for the synthesis of ATP from ADP and Pi;

Why these similarities exist

6. Both processes of ATP production are similar in the organelles because of the endosymbiont theory / endosymbiosis;
7. Mitochondria and chloroplasts originated as prokaryotic organisms which were taken inside a eukaryotic cell;

6 Max.

(c)

1. Under normal field conditions, carbon dioxide is the major limiting factor in photosynthesis, since its concentration in the atmosphere is about 0.03%.
2. Increasing carbon dioxide concentration leads to a linear increase until limited by other factors.
3. Ribulose bisphosphate carboxylase oxygenase (Rubisco), the enzyme that captures carbon dioxide in the light-independent reactions, has a binding affinity for both carbon dioxide and oxygen.
4. When the concentration of carbon dioxide is high, Rubisco will fix carbon dioxide in Calvin Cycle which increases the rate of photosynthesis.
5. If the carbon dioxide concentration is low and oxygen concentration is high, oxygen will out-compete carbon dioxide for the active site of the enzyme Rubisco during the dark stage of the reaction.
6. Therefore, a high concentration of oxygen lowers the rate of photosynthesis.